MINING THE STARS, MANAGING THE RISKS

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Executive Summary

Interstellar mining offers huge economic potential but carries extreme risks. Cosmic Quarry Mining operates across the Helionis Cluster, Bayesia System, and Oryn Delta, facing orbital debris, radiation surges, equipment stress, and hazardous workforce conditions. While routine losses are manageable, rare events like solar storms or systemic equipment failures can trigger extreme shocks, creating substantial financial losses. This report examines how Galaxy General Insurance can stabilise operations and strengthen financial resilience in this volatile environment.

1. Recommended Insurance Products and Strategic Rationale

Galaxy General will benefit from offering four primary products: Business Interruption (BI), Cargo Loss, Equipment Failure, and Workers' Compensation. Together, they address the most material exposures – operational downtime, mineral transport losses, infrastructure damage, and injuries.

- **Business Interruption:** Covers production disruptions from environmental hazards such as radiation spikes or debris impacts.
- **Equipment Failure:** Protects complex mining machinery in harsh space environments.
- **Cargo Loss:** Secures shipments of extracted minerals, with gold and platinum insured only via optional riders to limit concentration risk (see IIS, 2025).
- **Workers' Compensation:** Covers workforce injuries under hazardous conditions, supporting compliance and operational safety.

2. Product Design, System-Specific Tailoring, and Adaptability

Each product is modular and scalable to system-specific risks – Helionis Cluster faces high debris density, Bayesia System experiences radiation spikes and Oryn Delta contends with orbital shear and precious-metal logistics. Coverage limits, deductibles, and pricing adjust to operational intensity, equipment age, workforce size, and environmental risk with ESG incentives further enhancing enterprise risk management strategies outlined in the image below.

Figure 1 – Integration with Enterprise Risk Management (ERM) and Resilience

The outcome: Stronger operations, fewer shocks and resilient finances across the final frontier.

Product Design

Galaxy General's product design is scoped to the three solar systems previously mentioned in which Cosmic Quarry Mining Corporation operates. While the historical claims dataset spans five systems including Epsilon and Zeta, those systems are used only for model calibration and are not subject to coverage under this contract. Further details on model assumptions and product features are mentioned in *Appendix A*.

Feature	Cargo Loss (excl. gold + platinum)	Business Interruption	Workers' Compensation	Equipment Failure
a. Benefit basis	Indemnity of physical loss to cargo shipment at declared cargo value; sliding deductible: Đ50k (debris density< 0.6), Đ120k (0.6–0.8), Đ200k (> 0.8)	Indemnity of net revenue loss during the interruption period, capped at 180-day maximum indemnity period (MIP) per event	Wage replacement at 80% of base salary during disability period; lump-sum death/serious injury benefit scaled to base salary and occupation class	Reparation or replacement cost of insured unit at current replacement value, i.e. no betterment clause
b. Coverage triggers	Physical damage to cargo confirmed by telemetry; route risk tier ≥ 3 or debris density ≥ 0.6 activates higher deductible band; solar_radiation ≥ 0.7 activates radiation endorsement surcharge	Operational shutdown > 48-hour waiting period due to physical damage, supply chain severance (supply-chain index < 0.3), or radiation event; energy backup score ≤ 2 removes waiting period waiver	Confirmed workplace injury, illness, or fatality; occupational risk class confirmed; accident history lag and psych stress index used as rating variables, not trigger variables	Equipment failure confirmed by maintenance logs; equipment age, maintenance_int (100 – 5,000 hrs), and usage_int (0 – 24 hrs/day) are rating variables; failure during a declared radiation or debris event activates co-trigger with Cargo
c. Key exclusion	Gold and platinum shipments (base policy); pre-existing cargo damage; unapproved route deviations through debris_density > 0.9 corridors without relay telemetry	Losses from scheduled maintenance downtime; market price fluctuations unrelated to physical damage; events where production load was already ≤ 0 at policy inception	Injuries sustained outside declared working hours or off-site; pre-existing conditions not disclosed at underwriting; non-compliant protective gear (protective gear quality = 1)	Wear-and-tear deterioration (maintenance_int > 4,000 hrs without service); operator negligence where safety compliance = 1; equipment not listed in Cosmic Quarry inventory at policy inception
d. System-specific loading	Helionis: debris_density + route_risk primary; Bayesia: solar_radiation endorsement; Oryn Delta: catastrophe excess layer	Helionis: supply chain disruption primary; Bayesia: radiation shutdown; Oryn Delta: orbital-shear blackout	Solar system modelled as direct GLM rating factor in both frequency and severity models	Equipment type and solar system are primary categorical rating variables in both GLMs

Table 1 – Summary of Product Design by Hazard Coverage

Pricing and Profitability Insights

Across all hazard lines, premiums exceed expected losses and expenses (*Table 1 below & Appendix B*), supporting robust profitability. BI and Workers' Compensation exhibit low tail volatility, generating predictable cash flows and requiring relatively modest reserves. Equipment Failure presents moderate tail risk, justifying a higher risk loading to provide coverage for extreme events. While Cargo Loss drives portfolio tail exposure, excluding high-severity exposures (i.e. gold and platinum) materially reduces tail volatility whilst preserving underwriting profitability (Staudt, 2010).

All lines of business exhibit profit margins exceeding 10%, aligning with pricing targets. Reserve requirements vary by tail risk, as while BI and Workers' Compensation require relatively small reserves, Equipment Failure sits at an intermediate level and Cargo Loss retains higher reserves due to concentration risks. The present value of premiums comfortably covers expected losses, expenses and recommended reserves (see *Assumptions*), with 99% Tail Value-at-Risk (99% TVaR) reserves ensuring capital sufficiency for extreme but rare events. Portfolio diversification across all hazards smooths out capital requirements, mitigates aggregate tail risk, and enhances the predictability of cash flows

Summary of Pricing and Capital Modelling

Metric (Monte Carlo)	Business Interruption	Cargo Loss	Workers' Compensation	Equipment Failure
Expected Annual Loss	Đ 2.65 billion	Đ 15.38 billion	Đ 14.9 million	Đ 234.04 million
Standard Deviation	Đ 41.4 million	Đ 153.8 million	Đ 464,972.8	Đ 5.13 million
99% VaR	Đ 2.75 billion	Đ 15.70 billion	Đ 16.10 million	Đ 246 million
99% TVaR	Đ 2.76 billion	Đ 15.73 billion	Đ 16.28 million	Đ 248.13 million

Table 2 – Summary of Aggregate Loss Distributions

Aggregate loss distributions for each hazard coverage were estimated using Monte Carlo simulation, incorporating the calibrated frequency and severity models developed for each portfolio. The results (Table 2) indicate that BI represents the second largest exposure, with an expected annual loss of Đ2.65 billion and substantial tail risk. This is reflected in a 99% Value-at-Risk (VaR) of Đ2.75 billion and 99% TVaR of Đ2.76 billion, consistent with the systemic nature of operational disruptions.

Workers' Compensation exhibits moderate expected losses (Đ14.9 million), however demonstrates substantial tail exposure due to occasional high-severity claims, reflected in a 99% TVaR of Đ16.28 million. Although Equipment Failure and Cargo Loss display higher expected loss levels (Đ234.04 million and Đ15.38 billion, respectively), both portfolios remain exposed to adverse tail outcomes, demonstrated by their corresponding VaR and TVaR measures.

Stress Testing

Stress testing was undertaken across all four hazard coverages assessing the sensitivity of aggregate portfolio losses to extreme changes in key risk drivers and evaluating potential 1-in-100-year loss outcomes (99% VaR and TVaR). By examining the impact of these shocks on the

aggregate loss distribution (*see Appendix D*), primary contributors across all solar systems to tail risk within each coverage are identified, supporting the evaluation of underwriting controls.

Business Interruption

The results indicate that the BI portfolio is highly sensitive to shocks in claim frequency and severity, producing the largest increase in loss estimates under extreme scenarios. While baseline conditions produce losses of approximately Đ2.6– Đ2.7 billion, severe stress scenarios increase losses to over Đ5.4 billion, almost doubling insurers' tail risk exposure. Claim frequency and claim severity have the largest and most direct impact on extreme losses, highlighting that the portfolio is highly sensitive to changes in claim behaviour. However, operational risks such as production load and maintenance frequency produce similar increases in the 99% VaR under extreme stress, suggesting that operational disruptions within mining systems significantly amplify insurance losses. Another notable result is the impact of workforce factors, particularly safety compliance. Fluctuations in safety compliance levels result in leads to substantial changes to losses, indicating that human and organisational factors must be taken into consideration to control risk exposures. In contrast, claims inflation affects costs steadily, thereby increasing loss amounts at a lower rate than creating sudden extreme losses.

Cargo Loss

Stress Scenario	Mean Annual Loss	99% VaR	99% TVaR	99.5% VaR
Baseline	Đ15.38B	Đ15.70B	Đ15.73B	Đ15.73B
Frequency +50% (1-in-100)	Đ23.07B	Đ23.45B	Đ23.47B	Đ23.50B
Severity +50% (1-in-100)	Đ23.07B	Đ23.54B	Đ23.60B	Đ23.60B
Combined: Freq +30% & Sev +30%	Đ26.00B	Đ26.45B	Đ26.50B	Đ26.50B
Great Flare (Freq +50% & Sev +50%)	Đ34.61B	Đ35.19B	Đ35.26B	Đ35.26B

Table 3a – Cargo Loss Scenario Testing

Stress testing was conducted across eight risk drivers calibrated to all three operating environments (*Appendix D*). Under a 1-in-100 frequency shock, expected annual losses increase by 50% from Đ15.38 billion to Đ23.07 billion and the 99% VaR reaches Đ23.45 billion. An equivalent severity shock produces similar results, where the most severe scenario is a *Route Risk Surg*e of +35%. This captures the deterioration of the debris corridor and escalation of route-risk loading, increasing mean losses to Đ22.58 billion. The catastrophic "Great Flare" scenario, measured by modelling a correlated solar storm that simultaneously doubles the claim frequency and severity across all three solar systems, produces a mean loss of Đ34.61 billion and a 99% VaR of Đ35.19 billion, representing a 124% increase over the baseline. The 99.5% VaR under the Great Flare scenario of Đ35.26 billion defines Galaxy General's catastrophe capital loading for cargo.

Workers' Compensation

The Workers' Compensation portfolio shows moderate baseline losses but exhibits sensitivity to operational and organisational risk factors. Under baseline conditions, the portfolio generates a mean aggregate loss of Đ15.044 million, with a 99% VaR of Đ16.1 million *(Appendix D.3)*. Extreme stress scenarios indicate that claim frequency, claim duration and operational safety

conditions are the primary drivers of extreme losses. Increasing claim frequency by 50% increases losses by Đ7.6 million, while the tail risk only increases in similar proportion (up by 7.9 million). An equivalent severity shock produces similar impacts with a VaR increase of Đ8.1 million. Additional stress scenarios demonstrate that workplace conditions and environmental hazards significantly amplify tail outcomes. Elevated psychological stress levels double expected losses to Đ45.1 million with a VaR of Đ47.9 million, while severe environmental hazards (e.g., gravity-related operational risks) produce the most extreme outcomes, with the 99% VaR nearing Đ64.0 million.

Equipment Failure

The Equipment Failure portfolio demonstrates the greatest sensitivity to shocks in claim severity and compound operational assumptions, with the severity scale producing the widest VaR range across all stress scenarios. Severity scale spans approximately Đ85 million to over Đ200,000 million at the 99% VaR level, making it the dominant driver of tail risk, while equipment age and usage intensity follow closely behind (*Appendix D.4*). Frequency and maintenance intervals in isolation produce comparatively narrower ranges, confirming that single-driver frequency shocks are less destabilising than collective testing. The sensitivity line chart reinforces this, showing severity and compound assumptions exhibiting the steepest upward gradients as shocks intensify *(Appendix E.4)*. A combined 10% downward shock reduces mean loss from Đ5.1 million to Đ3.9 million (23%) and VaR from Đ131.4 million to Đ110.3 million (16%), hence reducing the capital buffer required. Conversely, a combined 10% upward shock elevates mean loss to Đ6.1 million (+19%) and VaR to Đ151.0 million (+15%), suggesting adverse deviations in operational conditions could place significant pressure on reserve adequacy and risk capital holdings.

Dependency – Based Methods

Cascading events can cause disproportionate losses, ones that no single hazard scenario could anticipate (SOA, 2025). Hence, to capture losses arising from events that simultaneously affect all three solar systems, such as a galaxy-wide radiation flare or supply chain collapse, a Gaussian copula was applied across all four hazards (*Appendix C*). Under independence ($\rho = 0$), the portfolio 1-in-40 VaR is approximately Đ22.5 billion, and the 1-in-200 VaR is Đ25.9 billion. Introducing a positive correlation of $\rho = +0.5$, representing the "Great Flare" scenario in which a solar storm simultaneously disrupts cargo transit, halts production, triggering business interruption, injures workers, and damages equipment across all three systems, it increases the 1-in-40 portfolio VaR by 1.6% to Đ22.9 billion and the 1-in-200 VaR by 2% to Đ26.4 billion. While the proportional increase appears modest, this reflects the dominance of Cargo Loss in the portfolio, making up over 85% of total VaR. Thus, the absolute additional capital requirement under the correlated stress scenario is approximately Đ530 million at the 1-in-200 level.

Risk Assessment

Risk Identification

Business Interruption

The Exploratory Data Analysis (EDA) highlights several drivers of BI risk across all solar systems. The hazard area coverage analysis indicates that exposure and frequency dominate the overall risk profiles (*Appendix E.1*), reflecting the dependence of BI losses on system capacity and operational utilisation. The average claim rate by exposure further suggests that claim rates are highest at very low exposure levels and decline as exposure increases, indicative of substantial volatility amongst

smaller operational units. However, at higher exposure levels, Epsilon and Zeta exhibit relatively higher claim rates compared with Helionis Cluster, suggesting that larger-scale operations in these systems may face greater disruption risk due to increased infrastructure complexity. The claim amount is strongly right-skewed, highlighting the presence of low-frequency but high-severity losses typical of BI portfolios. Additionally, its indicated that higher maintenance activity is associated with larger total claim amounts, particularly in Epsilon and Zeta, reflecting their greater operational exposure.

Cargo Loss

The Cargo Loss risk profile is characterised by two dominant physical drivers: debris density and route risk. EDA (Appendix E.2) reveals a near-monotonic increase in exposure-adjusted claim frequency across route risk tiers 1 to 5 and a similar gradient across debris density bands 0.0 to 1.0, with the highest-density band from 0.8–1.0, generating claim rates more than twice the lowest band (refer Appendix E). Solar radiation shows a secondary but meaningful effect, with the top radiation band 0.9–1.0 producing roughly 40% higher frequency than the base band. Cargo type is the dominant severity driver, as the excluded gold and platinum categories account for majority of total severity exposure. Among retained cargo types, lithium and cobalt carry the highest mean claim amounts, driven by their high per-unit cargo values and sensitivity to transit duration. Container type interacts significantly with both frequency and severity, with armoured containers showing lower claim rates but higher per-claim severity when losses occur.

Workers' Compensation

The workers' compensation risk profile is predominantly driven by occupation and the nature of injuries sustained (*Appendix E.3*). Drill operators, representing the largest workforce segment across all solar systems, consistently exhibit the highest claim frequency and severity, reflecting the inherently hazardous and physically intensive nature of their roles. In contrast, executive and administrative roles report lower claim counts, though individual claims in these groups are associated with higher per-claim costs due to wage-replacement exposures. Injury type further differentiates risk, with cuts, lacerations and sprains/strains being the most prevalent across high-frequency occupations, while stress-related injuries are more pronounced in specific locations, such as the Epsilon solar system, indicating operational or environmental contributors to claim incidence. Moreover, the claim severity distribution demonstrates a heavy right tail, capturing the potential for low-frequency, high-severity events that can materially impact the overall portfolio.

Equipment Failure

EDA findings reveal usage intensity, equipment type and solar system location as being the primary risk drivers (*Appendix D.4*). High intensity equipment have more frequent breakdowns and significant repair costs, thus contributing to the highest claim frequency and severity across all solar systems. Analysis of equipment types demonstrates FluxStream Carriers generating high claim volume regularly while Quantum Bore produces less frequent but larger losses, making these equipment types worthy of targeted pricing attention. Solar system location adds another layer of risk, with the Helionis Cluster consistently exhibiting the highest claim rates across all equipment types. Additionally, older equipment fails more often as claim frequency rises with equipment age before plateauing, suggests that policies covering ageing fleets may need to be reviewed more thoroughly at renewal.

Correlated Risk Scenarios

Correlation(rho)	Scenario	VaR 1-in-40 (97.5th)	Δ vs Indep (97.5th)	VaR 1-in-200 (99.5th)	Δ vs Indep (99.5th)
-0.5	Diversifying (negative corr.)	23,277	777	26,729	831
-0.2	Mildly diversifying	22,763	263	26,087	188
-0.1	Near independent (slightly negative)	22,614	114	25,859	-40
0.0	**Independence (baseline)**	**22,500**	**0**	**25,899**	**0**
+0.1	Mild positive correlation	22,727	227	26,303	404
+0.2	Moderate positive correlation	22,518	18	25,781	-118
+0.5	**High correlation — Great Flare scenario**	**22,858**	**358**	**26,428**	**529.3**

Table 3b – Summary of Correlated Risk Scenarios (Đ Millions)

To capture cross-hazard dependency, driven by common systemic shocks such as the "Great Flare" solar storm scenario, a Gaussian copula with 50,000 Monte Carlo simulations was applied across all four lines at correlation (ρ) levels ranging from -0.5 to 0.5 (refer *Appendix E*). Under independence, (ρ = 0), the 1-in-40 portfolio VaR is approximately Đ22.5 billion. under the high-correlation stress scenario, (ρ = 0.5), this rises to Đ22.9 billion, an increase of roughly 1.6%. In the 1-in-200 scenario, the VaR increases to Đ26.4 billion, illustrating that a correlated solar event meaningfully amplifies tail capital dependence across the portfolio and underscores the importance of correlated stress testing when pricing multi-line space insurance products.

Scenario Testing

Scenario testing evaluates the financial impact of varying operational environments across all four hazard coverages, using the simulated loss distribution and stress-testing results (*Appendix D*).

Scenario	Description	Annual Technical Premium	3-Year Expected Cost	3-Year Expected Net Revenue
Best Case	Attritional claims only; stable routes, low debris, no solar events	Đ20.3 billion	Đ37.4 billion	Đ50.9 billion
Moderate Case	Baseline pricing; moderate frequency and severity per GLM fit	Đ26.5 billion	Đ49.2 billion	Đ30.3 billion
Worst Case	Correlated catastrophe; Great Flare + debris cascade across all 3 systems	Đ55.6 billion	Đ115.9 billion	Đ23.4 billion

Table 4 – Scenario Testing Overview

Under the best-case scenario, characterised by attritional claims at below-average debris density and no radiation events, 3-year expected costs of Đ37.4 billion are comfortably covered by the Đ13.5 billion in projected returns, yielding a net revenue of Đ50.9 billion. The moderate case reflects the GLM-fitted baseline with a 3-year expected cost of Đ49.2 billion against Đ30.3 billion in net revenue. The worst case models a correlated catastrophe across all three systems simultaneously, driving 3-year costs to Đ115.9 billion. Net revenue of Đ23.4 billion indicates that there is a loss of Đ92.5 billion, and the 99th percentile cost tail reaches Đ744.0 billion over the 10-

year horizon, underscoring the need for adequate catastrophe capital reserving. Over a 10-year horizon under the worst case, the 99% VaR for costs reaches Đ2.56 trillion.

Threat Table

Rank	Threat Name	Rationale
1	Solar Storm / Correlated Flare	Drives the extreme losses in catastrophic modeling; affects all solar systems simultaneously.
2	Orbital Shear / Gravity Gradient	Asymmetric asteroid ring causes localised "crushing" events and high-severity equipment loss.
3	Radiation Spike / Binary Flare	Frequent K-type binary alignments cause "attritional" claims (high frequency, lower severity per claim).
4	Orbital Saturation (Debris)	Reflects the 2174 legal settlement (Đ 40M); manageable but poses a long-term liability risk.
5	Fatigue / Extended Hours Worked	Increases accident frequency, reflected in claim lengths
6	Climate Risk	Long-term environmental instability (solar cycles, asteroid drift, thermal fluctuations) disrupts infrastructure and increases systemic loss.
7	Inflation Risk	Rising repair, replacement, and wage costs increase claim severity, particularly for equipment replacement and workers' compensation.

The full impact and likelihood assessment of threats is highlighted in *Appendix G*.

Risk Differences

Helionis Cluster

Helionis Cluster has a high severity, moderate frequency risk profile across all lines. Workers' compensation claims are the costliest, driven by serious injuries including amputations and burns despite lower claim frequency. Cargo loss claim frequency is the highest due to dense metallic asteroid mining and debris collisions, though individual claim values are moderate. Comparatively, equipment failure rates are the lowest, suggesting a stronger maintenance discipline whilst BI losses show high variability, indicating elevated exposure to operational disruptions.

Bayesia System

Bayesia System's claim risk profile is shaped primarily by claim volume rather than claim severity. Workers' compensation frequency is slightly higher, with common occupational injuries contributing to greater losses. Equipment failure claim volumes are the highest of all three systems, particularly for Mag-Lift Aggregators and Quantum Bores, reflecting a more intensively operated fleet.

Oryn Delta

Oryn Delta mirrors Zeta's claim volumes and severity in most aspects. Operationally, elevated rates among higher-risk roles such as drill operators and engineers suggest workforce composition is a key loss driver. Equipment failure volumes are meaningful across Mag-Lift Aggregators and Quantum Bores, though consistently below Bayesia, pointing to either a smaller fleet or moderately better operational controls.

Environmental, Social and Governance (ESG) Features

Galaxy General's suite of insurance products for Cosmic Quarry exhibits multiple ESG dimensions, influencing both underwriting and long-term risk management.

Environmental: Environmental factors influence Business Interruption and Equipment Failure exposures through the operational characteristics of orbital mining. Energy-intensive extraction equipment contributes to emissions and resource depletion, which are implicitly reflected in risk-based loadings on downtime and equipment risks. Critical mineral cargo such as cobalt and lithium also faces supply-chain disruption linked to ESG pressures, including emissions, water stress and biodiversity impacts identified by the International Energy Agency (IEA, 2023). These risks are incorporated through adjusted BI and equipment-failure premiums, while mitigation tools such as telemetry-compliance programs and preventive maintenance reduce claim frequency.

Social: Worker safety is a key social factor across all lines. Cargo losses in Oryn Delta are correlated with worker safety incidents, as orbital-shear events that damage shipments frequently injure crew, creating a dependency between cargo, BI, and Workers' Compensation claims. ESG frameworks in mining consistently rank health and safety as the highest-priority social criterion (Fikru et al.). As a result, investments in safety protocols, crew training, and real-time monitoring reduce operational risk while supporting social responsibility. Reinvesting high-risk premium loadings from hazardous route tiers into telemetry-compliance programs can further reduce incidents while demonstrating commitment to ESG-focused investors.

Governance: Governance mechanisms strengthen risk management across all coverage lines. The Society of Actuaries notes that modern space insurance policies increasingly include collision-avoidance protocols (CRO, 2025), debris-mitigation requirements, and end-of-life disposal plans as regulatory and risk management tools. Galaxy General's debris-density triggers and route-change exclusions reinforce these practices by incentivising responsible operations. Maintenance schedules, automated fail-safes and reporting requirements further support operational discipline and regulatory compliance. As ESG compliance in mining expands globally, integrating ESG criteria into pricing helps position insurers to support evolving regulatory and reporting requirements.

Assumptions[1]:

The pricing and capital modeling framework relies on several key assumptions that directly influence premiums, reserve adequacy, and tail risk assessment. Frequency and severity distributions are calibrated to historical experience using GLM-derived models; cargo frequency follows a Negative Binomial distribution to capture overdispersion, while cargo severity is modeled with a heavy-tailed Gamma distribution. Workers' compensation and business interruption claims also reflect overdispersed frequency and skewed severity, anchored to observed portfolio losses to ensure alignment with historical total claims.

During data preparation, some variables contained additional symbols or suffixes appended to their names (e.g., "Drill operator" and "Drill operator_???123"). These were assumed to represent the same underlying variable and were consolidated accordingly to ensure consistency and avoid duplication in the modeling process.

[1] A comprehensive set of additional assumptions that are considered to have less material impact on pricing, capital modeling, and risk assessment are documented in Appendix H.

Economic assumptions include inflation-linked growth in claim costs and premiums (Appendix B.3), and the use of risk-free government bond yields for discounting future cash flows, ensuring realistic valuation of reserves and technical premiums. Dependence between risks and correlated catastrophic events are modeled via a Gaussian copula, with stress scenarios ($\rho = 0.1, 0.5, 0.9$) representing low, moderate, and extreme systemic correlations. Scenario multipliers for events such as the "Great Flare" adjust frequency and severity to assess multi-system losses. Monte Carlo simulation with 10,000 iterations provides stable tail risk estimates (e.g., 99.5th percentile VaR), while portfolio composition, underwriting practices, and profit targets are assumed stable over the projection horizon. Additional adjustments, such as the 20% co-insurance and Đ50M sublimit for precious metals shipments, mitigate concentration risk and moral hazard.

Data and Data Limitations:

Data Sources:

The analysis primarily uses the datasets provided in the *Project Data* section, including claims data for Cargo Loss, business interruption, workers' compensation, and equipment failure, along with Cosmic Quarry personnel and equipment inventory data for exposure estimation. A limited number of external sources were referenced for contextual support such as ESA (2025) for debris statistics, Lloyd's/Cambridge CRS (2024) for extreme space weather scenarios, IEA (2024) for ESG risks in mineral supply chains, and AIDA (2025) for governance considerations, but were used only to inform qualitative discussion and stress testing, not parameter estimation.

Data Limitations:

While the datasets provided were comprehensive for the purposes of this project, several limitations remain. Variables containing missing or inconsistent entries were removed during cleaning. However, this may have introduced selection bias, slightly skewing distributions. In addition, the severity dataset for Cargo Loss contains substantially fewer observations than the frequency dataset due to the exclusion of gold and platinum claims, meaning the severity model is estimated on a smaller subsample of claims. Cargo loss claims data also does not include explicit solar system identifiers, so system-level pricing relies on proxy variables such as route risk, debris density, and solar radiation rather than direct system tagging. Furthermore, for the solar systems, the historical data includes observations from the Epsilon and Zeta systems, which are outside the RFP scope and may dilute parameter estimates. The dataset is also cross-sectional and lacks a time dimension, preventing direct modelling of trends. As a result, extreme claim events may still be underrepresented and tail risk estimates may be conservative. These limitations demonstrate the usability of observations for understanding projected losses and stress testing, however, should be interpreted with caution, particularly for rare or catastrophic events. Furthermore, some of the parameter bounds given to us within the Data Dictionary, removed too many values when adhering to them, so were modified to only consider the lower bound for Workers Compensation.

Use of AI/LLM Tools:

AI tools were used to support coding checks, verify methodology, and assist with report formatting. All modelling, parameter estimation, and analysis were performed independently by the team and validated against the project datasets. AI therefore served as a supporting tool rather than a source of substantive analytical results. The team retains full responsibility for the actuarial judgement underlying assumption selection, model choices, and product design choices.

Appendix A – Full Rating Factor Structure

Appendix A.1– Cargo Loss Full Rating Factor Structure

Frequency model: Negative Binomial GLM, formula: claim_count ~ route_risk + pilot_experience + solar_radiation + debris_density + container_type + cargo_type + cargo_value + offset(log(exposure))

Severity model: Gamma GLM (log-link), formula: claim_amount ~ route_risk + solar_radiation + debris_density + cargo_value + weight + vessel_age + distance + transit_duration + pilot_experience + container_type + cargo_type

Variable	Type	Range / Levels	Direction (freq/sev)	Notes
route_risk	Continuous	1–5 (integer tiers)	+ / +	Primary pricing driver; tier 5 ~2× tier 1 frequency loading
debris_density	Continuous	0–1	+ / +	Also drives deductible band (0.6 / 0.8 thresholds)
solar_radiation	Continuous	0–1	+ / +	Activates radiation endorsement at ≥ 0.7
cargo_value	Continuous	Đ50k–Đ680M	+ / +	Direct severity multiplier; key concentration variable
weight	Continuous	1,500–250,000 units	+ / +	Correlated with cargo_value
pilot_experience	Continuous	1–30 years	− / −	Protective factor; reduces both frequency and severity
vessel_age	Continuous	1–50 years	+ / +	Older vessels have higher claim rates
distance	Continuous	1–100 AU	+ / +	Longer routes increase exposure duration
transit_duration	Continuous	1–60 days	+ / +	Correlated with distance; separate effect on severity

cargo_type	Categorical	Excl. gold/platinum; lithium, cobalt, iron ore, etc.	Varies	Gold/platinum excluded; lithium/cobalt carry highest severity loading
container_type	Categorical	Standard, armoured, pressurised, cryogenic	Varies	Armoured: lower frequency, higher severity

Deductible schedule:

Debris Density Band	Applicable Deductible
0.0 – 0.59	Đ50,000
0.60 – 0.79	Đ120,000
≥ 0.80	Đ200,000

Appendix A.2 – Business Interruption Full Rating Factor Structure

Frequency model: Negative Binomial GLM (log-link), formula: claim_count ~ solar_system + production_load + energy_backup_score + supply_chain_index + avg_crew_exp + maintenance_freq+ safety_compliance + offset(log(exposure))

Severity model: Gamma GLM (log-link), formula: claim_amount ~ solar_system + production_load + energy_backup_score + safety_compliance + exposure

Variable	Type	Range / Levels	Direction (freq/sev)	Notes
solar_system	Categorical	Helinois, Epsilon, Zeta	Varies	Captures environmental and operational differences across mining systems
production_load	Continuous	0 –1	+ / +	Higher production intensity increases equipment usage and operational strain
energy_backup_score	Ordinal	{0, 1, 2, 3, 4, 5}	− / −	Measures adequacy of emergency power systems; higher scores indicate stronger redundancy
supply_chain_index	Coninuous	0 – 1	– / –	Indicates stability of input sourcing; lower values imply

				supply disruption risk
avg_crew_exp	Continuous	1 – 30 Earth years	– / –	Higher workforce experience generally improves operational safety and response to disruptions
maintenance_freq	Continuous	0 – 6 cycles per year	− / −	More frequent maintenance reduces mechanical failure risk and lowers repair costs
safety_compliance	Ordinal	{1, 2, 3, 4, 5}	− / −	Compliance with safety protocols; higher scores correspond to safer operating environments
exposure	Continuous	0 – 1 Earth years	+ / 0	Exposure period used as an offset in frequency modeling

Deductible schedule:

Debris Density Band	Applicable Deductible
0.0 – 0.59	Đ70,000
0.60 – 0.79	Đ120,000
≥ 0.80	Đ200,000

Appendix A.3 – Workers' Compensation Full Rating Factor Structure

Frequency model: Negative Binomial GLM (log-link), formula: claim_count ~ occupation + solar_system + accident_flag_history + psych_stress_index + safety_training_index + offset(log(exposure))

Severity model: Lognormal GLM (identity link), formula: log(claim_amount) ~ occupation + solar_system + employment_type + accident_history_flag + psych_stress_index + supervision_level + protective_gear_quality+ base_salary + claim_length

Variable	Type	Range / Levels	Direction (freq/sev)	Notes

solar_system	Categorical	Helinois, Epsilon, Zeta	Varies	Captures environmental and operational differences across mining systems
occupation	Categorical	Administrator, Manager, Executive, Scientist, Engineer, Drill Operator	+ / +	Operational roles (e.g. Drill Operator) have higher accident exposure
employment_type	Categorical	Full-time, Part-time, Contract	+ / +	Contract workers face higher risks due to shorter training cycles
accident_history_flag	Binary	{0,1}	+ / +	Workers with prior injuries have higher future claim probability
psych_stress_index	Ordinal	1-5 (tier index)	+ / +	Higher stress levels associated with increased accident likelihood
supervision_level	Continuous	0 – 1 (tier index)	− / −	Higher supervisor-to-worker ratios improve safety and reduce injury severity
safety_training_index	Ordinal	1-5 (tier index)	− / −	Better training reduces accident frequency; smaller effect on severity
protective_gear_quality	Ordinal	1 – 5 (tier index)	~ / −	Better equipment reduces injury severity
exposure	Continuous	0 – 1 years	+ / −	Used as an exposure offset in the frequency model to scale claim counts

base_salary	Continuous	20K – 130K	~ / +	Higher wages increase compensation payments
claim_length	Continuous	3 – 1000 days	~ / +	Longer recovery periods increase total claim costs

Appendix A.4 – Equipment Failure Full Rating Factor Structure

Frequency model: Poisson GLM (log-link), formula: claim_count ~ equipment_age + solar_system + maintenance_int + usage_int + offset(log(exposure))

Severity model: Lognormal GLM (identity link), formula: log(claim_amount) ~ equipment_type + equipment_age + solar_system + maintenance_int + uasage_int

Variable	Type	Range / Levels	Direction (freq/sev)	Notes
equipment_type	Categorical	{Quantum Bore, Graviton Extractor, FluxStream Carrier, Mag-Lift Aggregator, Fusion Transport, Ion Pulverizer}	Varies	Captures structural differences in machinery design and operating environment
solar_system	Categorical	Helinois, Epsilon, Zeta	Varies	Captures environmental and operational differences across mining systems
equipment_age	Continuous	0 – 10 Earth years	+ / +	Aging equipment exhibits higher mechanical wear
maintenance_int	Continuous	100 – 5,000 Earth hours	+ / +	Longer intervals between maintenance increase failure likelihood
usage_int	Continuous	0 – 24 Earth hours per day	+ / +	Higher daily operating time increases mechanical stress and utilisation

exposure	Continuous	0 – 1 (tier index)	+ / 0	Exposure period used as an offset in the frequency model

Appendix B – Pricing and Profitability Insights

Appendix B.1 – Present Value of Cash Flows

Line of Business	PV Premiums (Đ)	PV Losses (Đ)	PV Expenses (Đ)	PV Profit (Đ)
Business Interruption	41,009,976,589	24,619,395,729	12,302,992,977	4,087,587,883
Cargo Loss (no gold/platinum)	2,379,777,125,71	1,427,866,275,43	713,933,137,771	237,977,712,571
Equipment Failure	39,056,621,651	23,561,105,651	11,716,986,491	3,778,529,501
Workers Compensation	230,704,191	138,422,515	69,211,257	23,070,419
TOTAL	2,831,240,555,16	1,699,005,563,52	849,372,166,654	282,862,825,091

Table 4.1 – Enhanced Summary of Metrics by Hazard Coverage

Line of Business	Recommended Reserve (Đ)	Tail Gap 99 (Đ)	Profit Margin	Reserve Proportion
Business Interruption	2,760,838,551	108,621,033	10%	15%
Cargo Loss (no gold/platinum)	159,200,243,19	37,794,833	10%	84%
Equipment Failure	2,771,227,421	23,301,820	9.7%	1.5%
Workers Compensation	18,432,317	350,994	10%	0.1%
TOTAL	189,764,179,29	N/A	10%	1.00

Table 4.2 – Enhanced Summary of Metrics by Hazard Coverage

Appendix B.2 – Cashflow Projections

Đ Millions	2175	2176	2177	2178	2179
premium_forecast	430.36	440.54	451.54	463.14	475.04
loss_forecast	259.61	265.76	272.39	279.39	286.57
operating_expenses	(129.11)	(132.16)	(135.46)	(138.94)	(142.51)
operating_income	125.34	128.30	131.51	134.88	138.35
net_income	**92.87**	**95.07**	**97.45**	**99.95**	**102.52**
pv_premium	410.44	403.24	398.13	394.23	390.88
pv_loss	(247.60)	(243.25)	(240.17)	(237.82)	(235.80)
pv_expense	(123.13)	(120.97)	(119.44)	(118.27)	(117.26)
pv_net_income	**88.58**	**87.02**	**85.92**	**85.08**	**84.35**

Đ Millions	2180	2181	2182	2173	2174
premium_forecast	487.10	499.32	511.78	524.58	537.76
loss_forecast	293.85	301.22	308.74	316.46	324.40
operating_expenses	(146.13)	(149.80)	(153.54)	(157.38)	(161.33)
operating_income	141.86	145.42	149.05	152.78	156.62
net_income	**105.12**	**107.76**	**110.45**	**113.21**	**116.05**
pv_premium	387.72	384.65	381.66	378.77	375.96
pv_loss	233.90	232.04	230.24	228.49	226.80
pv_expense	116.32	115.39	114.50	113.63	112.79
pv_net_income	**83.67**	**83.01**	**82.36**	**81.74**	**81.14**

Appendix B.3 – Cashflow Calculation Methodology

1. Historical financial data for each line of business over the past three years was collected from internal accounting records to support the financial and actuarial analysis. Key metrics extracted included net revenue, operating expenses, operating income, other income, pre-tax income, income tax, and net income. The data was cleaned and normalised to ensure consistent naming conventions and formatting across financial line items, allowing accurate aggregation and analysis across different business segments.

2. Operating expenses and profit loadings were incorporated using fixed ratios applied to the technical premium. An expense ratio of 30% and a profit margin of 7% were assumed based on historical insurer operating structures and industry benchmarks. These ratios were applied consistently across the pricing model to translate pure premiums into technical premiums while maintaining a stable and transparent pricing framework.

3. Macroeconomic factors were incorporated into the projections through inflation and risk-free interest rate forecasting. Historical inflation and risk-free rate data, including overnight bank lending rates and 1-year and 10-year government bond yields, were modelled as separate time series. ARIMA models were fitted to each series to produce ten-year forecasts, capturing expected economic trends. These forecasts were used to adjust projected premiums, claim costs, and discount rates used in present value calculations.

4. Claim frequency and severity were modelled separately using a standard actuarial frequency-severity framework with each line of business using a unique model. The models were selected based on model selection criteria including AIC, BIC and Vuong test diagnostics. Extreme losses were also captured in various ways (mostly Gamma distribution with optional Generalised Pareto Distribution) as it provides a realistic representation of rare but high-severity claims affecting portfolio risk. After this, aggregate portfolio losses were simulated using Monte Carlo methods where for each iteration, claim counts were drawn from the fitted frequency model and claim severities from the severity distribution.

5. Baseline pre premiums were calculated as the product of expected claims frequency and expected claim severity for each line of business. A risk margin of 5% of the pure premium was applied to account for uncertainty. Technical premiums were then calculated using the following formula:
   Technical Premium = (Pure Premium + Risk Margin) / (1 – Expense Ratio – Profit Ratio)

6. Reserve requirements were determined using a risk-based metrics derived from the simulated portfolio loss distribution. Key measure included Value-at-Risk (VaR) at the 95% and 99% levels and Tail Value-at-Risk (TVaR) at the 99% level. Recommended reserves for each line of business were defined as the maximum of expected loss plus TVaR margin, VaR 99% or TVaR 99%, ensuring adequate capital to cover rare but sever loss events.

7. Projected results were aggregated across all lines of business to produce consolidated portfolio projections. The framework assumes independence between lines beyond historical experience and includes ten-year financial projections, present value estimates of premiums, losses, expenses, and net income, and recommended reserves for each line and the overall portfolio. This methodology integrates histocial performance, macroeconomic forecasts, stochastic modelling, and risk-based capital assessment to support robust evaluation of pricing adequacy, financial sustainability, and capital requirements across the portfolio.

Appendix B.3 – Interest and Inflation

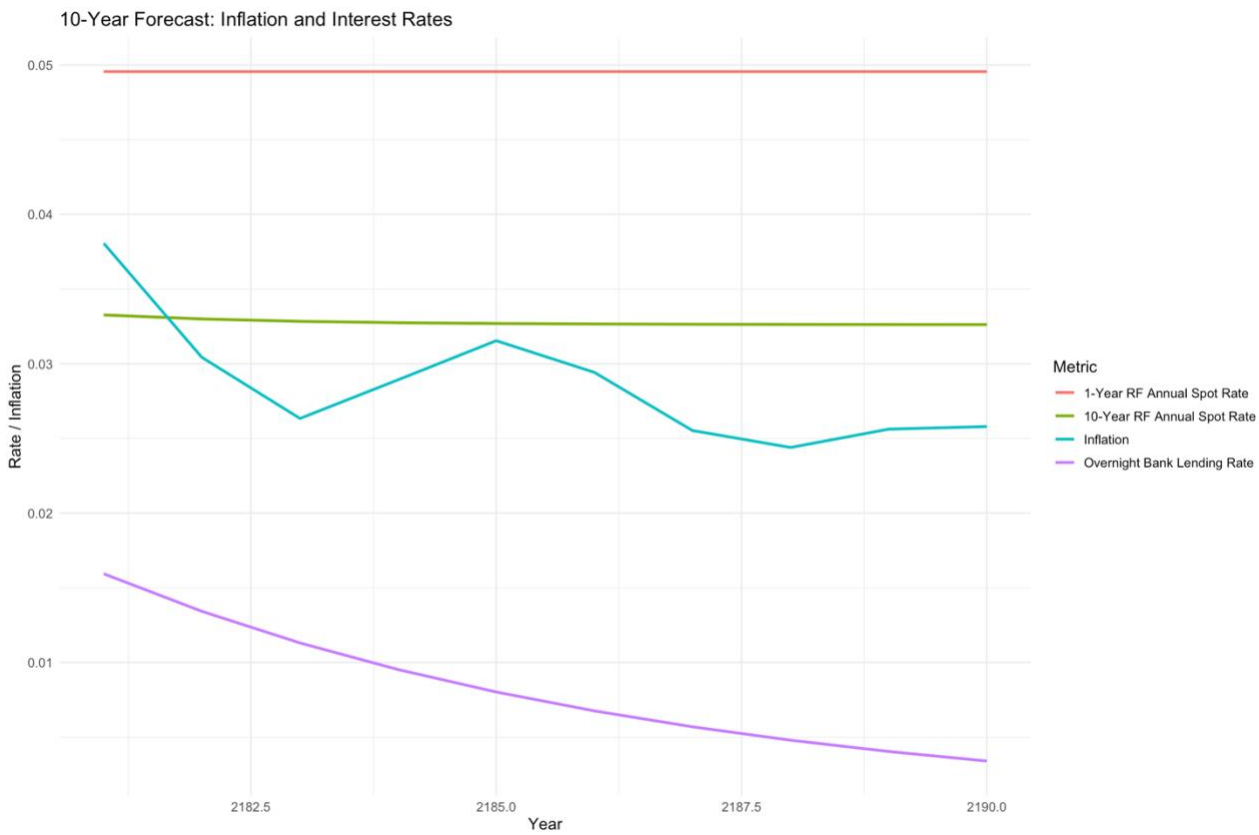


Figure 3 – Interest and Inflation Forecast

To project future inflation and interest rate conditions relevant to pricing, an ARIMA (Autoregressive Integrated Moving Average) model was fitted independently to each of the four economic series (inflation, overnight bank lending rate, one-year risk-free annual spot rate, and ten-year risk-free annual spot rate) using historical data provided in the project dataset. The ARIMA framework was selected for its flexibility in capturing autocorrelation structures and non-stationary trends commonly exhibited in macroeconomic time series. Each model order was selected automatically via maximum likelihood optimization, and a ten-year forecast horizon was generated for each series. The resulting projections, illustrated in the above figure, inform the discount rate and loss trend assumptions embedded in the pricing model, ensuring that expected future costs and returns are appropriately adjusted for the time value of money and interplanetary inflationary pressures across the Helionis Cluster, Bayesia System, and Oryn Delta solar systems.

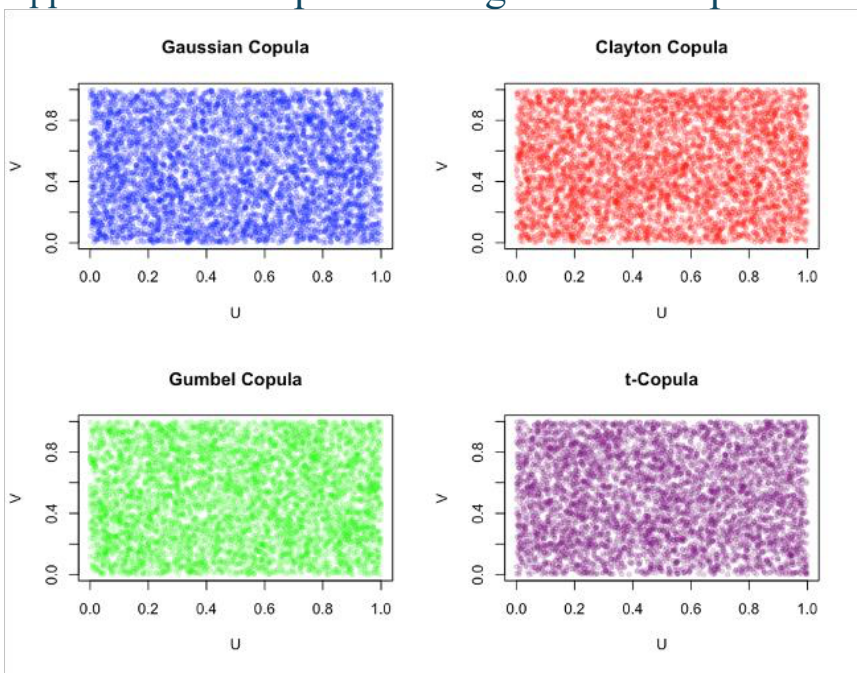
```
library(forecast)
library(tsibble)
library(dplyr)
library(fable)
library(feasts)
library(readxl)
library(ggplot2)
library(tidyr)

# Load Excel
```

```
inflation <- read_excel("srcsc-2026-interest-and-inflation.xlsx")
colnames(inflation) <- as.character(inflation[2, ])
inflation <- inflation[-c(1,2), ]

# Convert numeric
inflation$Year <- as.numeric(inflation$Year)
inflation$Inflation <- as.numeric(inflation$Inflation)
inflation$`Overnight Bank Lending Rate` <- as.numeric(inflation$`Overnight Bank Lending
Rate`)
inflation$`1-Year RF Annual Spot Rate` <- as.numeric(inflation$`1-Year Risk Free Annual
Spot Rate`)
inflation$`10-Year RF Annual Spot Rate` <- as.numeric(inflation$`10-Year Risk Free Annual
Spot Rate`)

# Reshape to long format
infl_long <- inflation %>%
  select(Year, Inflation, `Overnight Bank Lending Rate`, `1-Year RF Annual Spot Rate`, `10-
Year RF Annual Spot Rate`) %>%
  pivot_longer(-Year, names_to = "Metric", values_to = "Value")

# Convert to tsibble
infl_ts <- infl_long %>%
  as_tsibble(key = Metric, index = Year)

# Fit ARIMA model for each series
infl_models <- infl_ts %>%
  model(ARIMA(Value))

# Forecast 10 years ahead
infl_forecasts <- infl_models %>%
  forecast(h = 10)

# Extract forecast data as a data.frame for plotting
forecast_df <- infl_forecasts %>%
  as_tibble() %>%
  select(Year, Metric, .mean)

# Plot
ggplot(forecast_df, aes(x = Year, y = .mean, color = Metric)) +
  geom_line(linewidth = 1) +
  labs(
    title = "10-Year Forecast: Inflation and Interest Rates",
    x = "Year",
    y = "Rate / Inflation",
    color = "Metric"
  ) +
```

```
  theme_minimal() +
  theme(text = element_text(size = 12))
```

Appendix C – Dependency-Based Methods

Appendix C.1– Copula Testing for Tail-Dependence



To examine potential dependence between claim frequency and severity for each hazard, several copula structures were evaluated, including the Gaussian, Clayton, Gumbel, and t-copula. The simulated pseudo-observations from each fitted copula for cargo loss are shown in the figure above, though all hazards displayed similar results. Across all four copula structures, the simulated points are broadly dispersed across the unit square without strong clustering in the upper-right or lower-left regions, indicating limited evidence of strong dependence between claim frequencies and claim severities in the observed data.

Appendix C.2– Total Portfolio Aggregate Loss by Correlation Level (Gaussian Copula)

Total Portfolio Aggregate Loss by Correlation Level (Gaussian Copula)									
Gaussian copula across all 4 hazard lines | rho = 0 is pure independence | rho = +0.5 models correlated solar storm / Great Flare scenario									
Correlation (rho)	**Scenario Label**	**Mean Loss**	**Std Dev**	**VaR 1-in-20 (95th)**	**TVaR 1-in-20 (95th)**	**VaR 1-in-40 (97.5th)**	**TVaR 1-in-40 (97.5th)**	**VaR 1-in-200 (99.5th)**	**TVaR 1-in-200 (99.5th)**

-0.5	**Diversifying (negative corr.)**	17,623,640,029	2,013,852,781	21,605,344,659	23,865,126,634	23,277,390,522	25,390,397,745	26,729,480,446	28,814,685,201
-0.2	**Mildly diversifying**	17,546,156,801	1,884,388,814	21,165,878,251	23,290,843,077	22,763,244,790	24,724,899,367	26,087,201,282	27,663,177,165
-0.1	**Near independent (slight correlation)**	17,528,434,956	1,869,072,406	21,180,402,904	23,195,718,894	22,614,143,604	24,593,464,765	25,858,604,308	27,443,643,171
0.0	**Independence (baseline)**	**17,526,860,070**	**1,891,770,437**	**21,196,705,392**	**23,167,015,601**	**22,499,979,429**	**24,556,556,764**	**25,898,825,106**	**27,776,389,322**
+0.1	**Mild positive correlation**	17,525,693,769	1,959,001,882	21,255,941,657	23,416,790,472	22,726,824,099	24,937,724,833	26,302,585,576	28,281,929,275
+0.2	**Moderate positive correlation**	17,498,442,965	1,923,102,452	21,179,441,193	23,143,405,832	22,518,047,720	24,527,712,553	25,781,080,679	27,912,337,063
+0.5	**High correlation - Great Flare scenario**	**17,532,817,477**	**2,012,424,300**	**21,442,728,112**	**23,503,064,286**	**22,857,577,806**	**24,905,507,403**	**26,428,172,551**	**28,029,881,034**

Appendix D – Stress Testing

Appendix D.1– Business Interruption Stress Testing

Risk Factor	Stress Factor	Mean Loss	VaR (99%)
CORE INSURANCE RISK			
Claim Frequency	1.00	2,683,406,118	2,707,869,109
Claim Frequency	1.25	3,243,670,760	3,267,541,255
Claim Frequency	1.50	3,987,755,273	4,013,548,480
Claim Frequency	1.75	4,601,213,621	4,629,502,968
Claim Frequency	2.00	5,395,544,023	5,410,982,712
Claim Severity	1.00	2,699,093,586	2,713,844,520
Claim Severity	1.25	3,335,632,232	3,354,943,177
Claim Severity	1.50	3,972,420,984	4,001,037,786
Claim Severity	1.75	4,693,034,762	4,699,597,714
Claim Severity	2.00	5,276,132,446	5,298,236,412
OPERATIONAL RISK			

Production Load	1.00	2,639,488,156	2,655,333,577
Production Load	1.10	2,910,666,578	2,917,590,809
Production Load	1.25	3,298,600,090	3,320,335,768
Production Load	1.50	3,987,456,303	4,014,939,112
Production Load	1.75	4,588,882,780	4,605,261,394
Production Load	2.00	5,452,376,838	5,470,329,945
Energy Backup Score	1.00	2,558,142,859	2,565,632,233
Energy Backup Score	1.10	2,891,931,839	2,909,446,207
Energy Backup Score	1.25	3,288,424,364	3,300,847,576
Energy Backup Score	1.50	4,029,180,486	4,038,845,557
Energy Backup Score	1.75	4,701,631,167	4,709,662,285
Energy Backup Score	2.00	5,323,989,294	5,344,899,067
Supply Chain Index	1.00	2,677,096,171	2,694,148,542
Supply Chain Index	1.10	2,917,722,221	2,926,313,953
Supply Chain Index	1.25	3,340,614,255	3,357,442,479
Supply Chain Index	1.50	3,936,909,565	3,957,769,038
Supply Chain Index	1.75	4,781,994,903	4,810,985,019
Supply Chain Index	2.00	5,269,205,640	5,293,613,628
Maintenance Frequency	1.00	2,669,536,983	2,679,855,806
Maintenance Frequency	1.10	2,923,421,357	2,930,320,425
Maintenance Frequency	1.25	3,301,849,697	3,322,520,220
Maintenance Frequency	1.50	3,880,841,900	3,901,690,038
Maintenance Frequency	1.75	4,615,547,858	4,624,226,946
Maintenance Frequency	2.00	5,446,060,984	5,462,265,299
WORKFORCE RISK			
Crew Experience	0.60	1,624,820,827	1,637,903,783
Crew Experience	0.80	2,170,013,634	2,193,573,602
Crew Experience	1.00	2,680,943,216	2,698,766,764
Crew Experience	1.20	3,189,379,126	3,214,890,052
Crew Experience	1.40	3,701,230,785	3,715,660,940
Safety Compliance	0.80	2,137,030,605	2,147,736,091
Safety Compliance	1.00	2,645,524,924	2,658,757,686
Safety Compliance	1.20	3,173,494,672	3,184,533,190
Safety Compliance	1.40	3,755,366,148	3,785,121,266
Safety Compliance	1.60	4,250,107,371	4,293,579,701
PORTFOLIO RISK			
Exposure Growth	1.00	2,598,565,563	2,608,876,056
Exposure Growth	1.25	3,402,611,877	3,420,919,920
Exposure Growth	1.50	4,087,174,187	4,097,096,313
Exposure Growth	1.75	4,696,692,565	4,720,745,583
Exposure Growth	2.00	5,316,555,273	5,321,700,964

FINANCIAL RISK			
Claims Inflation	1.00	2,623,657,606	2,642,037,238
Claims Inflation	1.05	2,748,369,819	2,750,461,314
Claims Inflation	1.10	2,923,188,835	2,952,338,497
Claims Inflation	1.15	3,202,357,463	3,215,438,794
Claims Inflation	1.20	3,156,746,287	3,172,737,814
Claims Inflation	1.30	3,465,543,307	3,479,049,161
Claims Inflation	1.40	3,620,737,723	3,635,561,010
Claims Inflation	1.50	3,983,219,700	3,996,241,416

Appendix D.2– Cargo Loss Stress Testing

Stress Test	Stress Factor	Mean Loss (Đ)	VaR 99% (Đ)
Frequency	0.85	13074628641.4928	13482837777.1742
Frequency	1	15382852850.3703	15846621632.2599
Frequency	1.15	17689799748.4582	18210243266.7061
Frequency	1.3	19996650930.7916	20573386763.7609
Frequency	1.5	23072129617.2541	23715884875.3262
Severity	0.85	13075149297.8298	13473171205.0346
Severity	1	15382531848.1855	15855747813.4493
Severity	1.15	17686884445.2286	18226650498.2185
Severity	1.3	19998285071.0852	20601321780.9417
Severity	1.5	23071696531.9922	23786368943.0856
Route Risk	0.9	13499372288.8571	13917517629.8016
Route Risk	1	15381270053.3803	15849777610.2414
Route Risk	1.1	17343104177.0113	17866721433.2871
Route Risk	1.2	19382413192.87	19948010977.7137
Route Risk	1.35	22583228259.4512	23236590848.2852
Solar Radiation	0.85	14136822563.9777	14580297127.5619
Solar Radiation	1	15383242240.5705	15852400919.4119
Solar Radiation	1.15	16677940069.6273	17177949371.9219
Solar Radiation	1.3	18017557729.3006	18560222373.2761

Solar Radiation	1.45	19406912505.2202	19975884634.8052
Debris Density	0.85	13806584680.3265	14229144243.0956
Debris Density	1	15382591020.5321	15852698908.5738
Debris Density	1.15	17036323405.0624	17543579764.8769
Debris Density	1.3	18769404558.664	19332763123.1116
Debris Density	1.45	20577121815.8241	21188881382.8553
Claims Inflation	1	15381122375.5413	15856007579.91
Claims Inflation	1.05	16151207318.8116	16641012623.5777
Claims Inflation	1.1	16918747742.2556	17437632737.3157
Claims Inflation	1.2	18457147736.7332	19020041881.4503
Claims Inflation	1.3	19996003480.5959	20613401062.7799
Claims Inflation	1.4	21532694401.722	22184918657.1038
Freq-Sev Dependence	0	15282699520.1925	90706347840.1427
Freq-Sev Dependence	0.2	15460774251.266	92560344118.9511
Freq-Sev Dependence	0.4	15488032927.3535	94059935291.1327
Freq-Sev Dependence	0.6	15315924821.341	92965085866.8808
Freq-Sev Dependence	0.8	15753432102.3274	96183200942.4463
Catastrophe Probability	0	15490458093.7628	93076052576.533
Catastrophe Probability	0.005	15912062509.1559	97719517130.9383
Catastrophe Probability	0.01	16251461681.5897	101508272916.108
Catastrophe Probability	0.02	17011724355.2202	111194027792.791
Catastrophe Probability	0.03	17650995175.7264	124620543099.576

Appendix D.3– Workers' Compensation Stress Testing

Core Portfolio Risk Divers

Stress Test	Stress Factor	Mean Loss (Đ)	VaR 99% (Đ)
Baseline	1.0	15,044,378	16,110,536
Frequency	1.25	18,799,709	20,085,865
Frequency	1.50	22,568,856	24,040,225
Frequency	1.75	26,315,790	28,011,776
Frequency	2.0	30,080,631	32,001,203
Severity	1.25	18,808,653	20,167,757
Severity	1.50	22,560,708	24,204,903
Severity	2.00	30,092,724	32,256,935
Claim length	1.25	18,796,994	22,093,986
Claim length	1.50	22,591,277	26,530,208
Claim length	2.00	30,112,024	35,377,406

Operational Risk Drivers

Risk Factor	Stress Factor	Mean Loss (Đ)	VaR 99% (Đ)
Exposure	1.50	22,560,279	24,089,822
Hours Worked	2.00	30,089,817	31,959,482
Hours Worked	2.50	37,611,211	39,965,493
Supervision (Poor)	0.4	24,072,166	25,656,799
Experience (Low)	0.4	24,058,802	25,638,205

Safety and Environment Drivers

Risk Factor	Stress Factor	Mean Loss (Đ)	VaR 99% (Đ)
Training / PPE	0.50	7,518,168	8,057,192
Training / PPE	2.00	30,090,539	32,270,444
Gravity Hazard	2.00	60,178,558	64,015,228

Behavioural & Financial Drivers

Risk Factor	Stress Factor	Mean Loss (Đ)	VaR 99% (Đ)
Psychological Stress	2.00	45,126,746	47,927,100
Claims inflation	1.50	22,569,080	24,156,172
Salary Linked	1.50	22,560,384	24,169,966

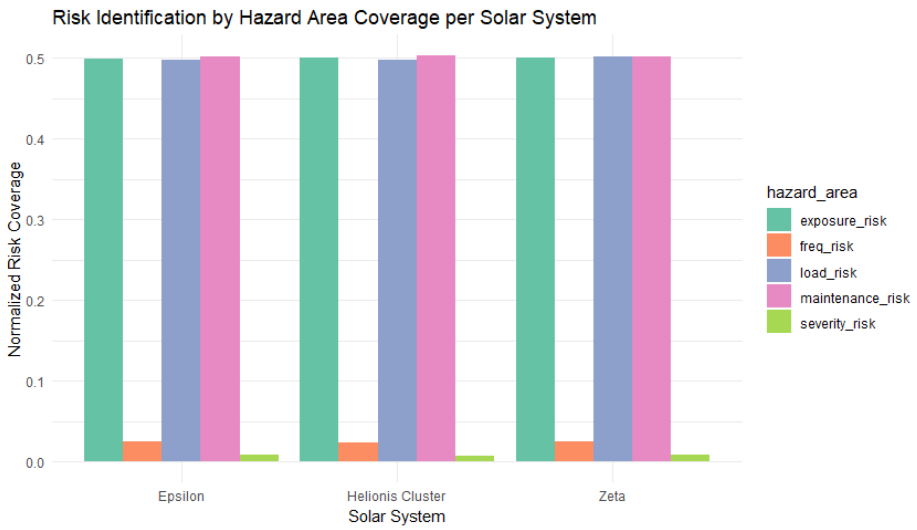
Appendix D.4– Equipment Failure Stress Testing

Assumption	Shock	Mean Loss	VaR 99%	TVaR 99%
Baseline	0%	5116	131412	172592
Frequency (lambda)	-50%	2655	100890	148496
Frequency (lambda)	-40%	2963	106354	140852
Frequency (lambda)	-30%	3608	117558	154700
Frequency (lambda)	-20%	3949	120386	163024
Frequency (lambda)	-10%	4638	125662	167881
Frequency (lambda)	10%	5666	132974	181555
Frequency (lambda)	20%	5974	133837	173321
Frequency (lambda)	30%	6453	143771	194584
Frequency (lambda)	40%	6749	145346	189334
Frequency (lambda)	50%	7753	149761	196731
Severity (scale)	-50%	2520	63893	86006
Severity (scale)	-40%	2883	72956	99976
Severity (scale)	-30%	3484	91177	123032
Severity (scale)	-20%	4046	104458	136886
Severity (scale)	-10%	4613	120797	155837
Severity (scale)	10%	5476	138002	183934
Severity (scale)	20%	5963	157784	203737
Severity (scale)	30%	6454	167229	214172
Severity (scale)	40%	6910	180959	238746
Severity (scale)	50%	7479	198633	258483
Maintenance Interval	-50%	2524	95263	137243
Maintenance Interval	-40%	3006	104347	141923
Maintenance Interval	-30%	3971	120255	165543
Maintenance Interval	-20%	4228	118600	165753

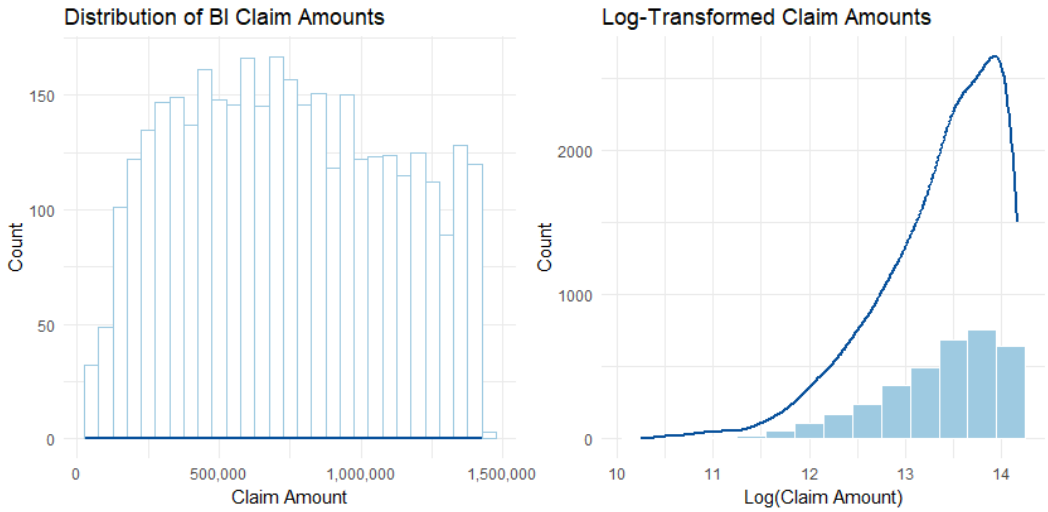
Maintenance Interval	-10%	4757	129254	168011
Maintenance Interval	10%	5353	133464	169269
Maintenance Interval	20%	5921	136744	180696
Maintenance Interval	30%	6913	143978	189427
Maintenance Interval	40%	6518	139389	180038
Maintenance Interval	50%	7588	143770	193886
Usage Intensity	-50%	1769	66621	100008
Usage Intensity	-40%	2694	92211	121049
Usage Intensity	-30%	3068	95306	143909
Usage Intensity	-20%	3676	104935	143297
Usage Intensity	-10%	3774	110265	150758
Usage Intensity	10%	5676	142574	182606
Usage Intensity	20%	6649	151405	194687
Usage Intensity	30%	7534	159364	216471
Usage Intensity	40%	8398	173601	232330
Usage Intensity	50%	9687	182201	236549
Equipment Age	-50%	1902	71940	107555
Equipment Age	-40%	2519	84616	117966
Equipment Age	-30%	3023	97578	129793
Equipment Age	-20%	3333	101557	135766
Equipment Age	-10%	4316	119577	155620
Equipment Age	10%	6101	143836	185330
Equipment Age	20%	6686	150520	201575
Equipment Age	30%	7830	168861	215811
Equipment Age	40%	8418	179220	230279
Equipment Age	50%	10048	191931	238120

Appendix E – Exploratory Data Analysis (EDA)

Appendix E.1– Business Interruption EDA

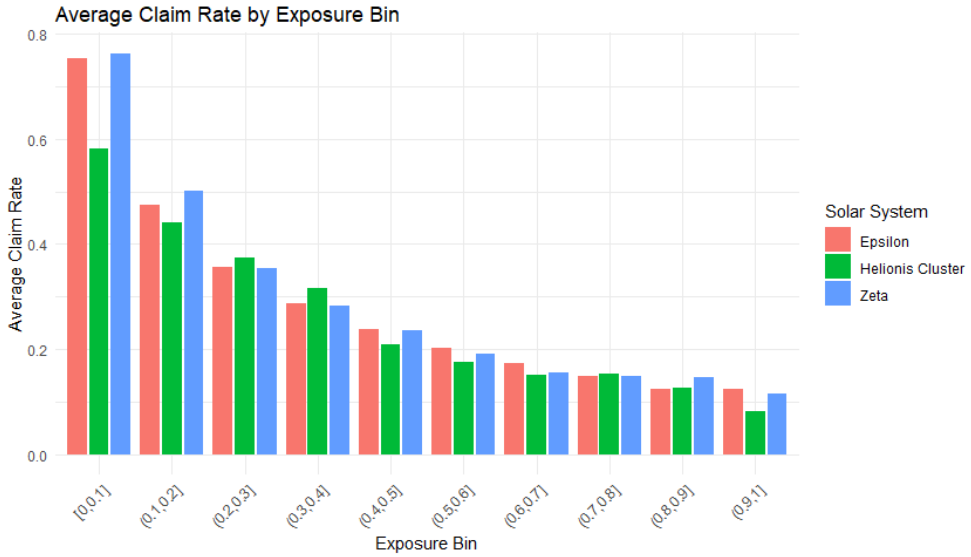


In the figure above, the risk identification by hazard area coverage across the three solar systems, Epsilon, Helionis Cluster, and Zeta, reveals a broadly consistent risk profile, with most of the normalized risk coverage dominated by exposure_risk, load_risk, and maintenance_risk across all systems. Notably, severity_risk contributes marginally low coverage values, suggesting that while catastrophic risk events are present, they represent a disproportionately small share of total hazard area coverage. The Helionis Cluster exhibits a marginally more balanced distribution across hazard types compared to Epsilon and Zeta, which display near-identical risk compositions, indicating that geographic or structural differences between systems have limited influence on overall risk allocation.

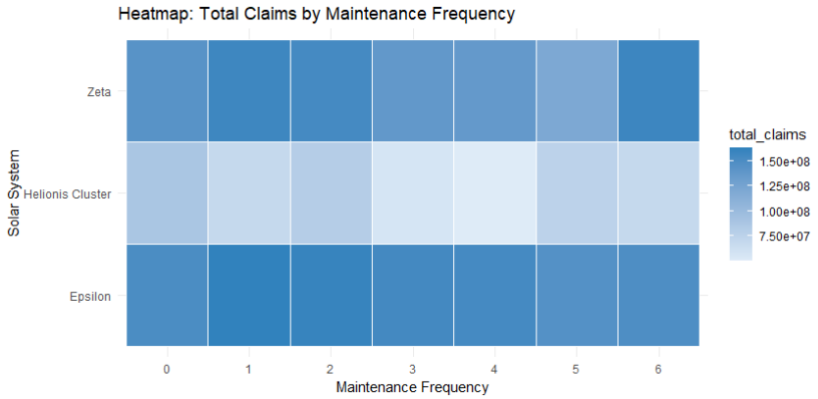


In the figure above, the distribution of Business Interruption (BI) claim amounts appears relatively uniform across a wide range, with counts consistently between 100 and 165 for most bins spanning from around Đ200,000 to Đ1,000,000, before gradually tapering off at higher values. This relatively flat distribution with a slight drop-off in the upper tail suggests that BI claims are broadly spread across a wide range of severities rather than concentrated at lower values, which is atypical compared to most claim distributions. A log transformation was applied to better reveal the underlying shape of the distribution, which exposes a strong right skew in the

log scale, with the density curve peaking sharply around 14 and rapidly declining on either side. This confirms that while moderate-to-large claims dominate the portfolio, extremely large claims remain rare. The wide spread of raw claim amounts, reaching up to approximately Đ1,500,000, highlights the significant variability in BI claim severity and suggests that tail risk management is an important consideration for this line of business.

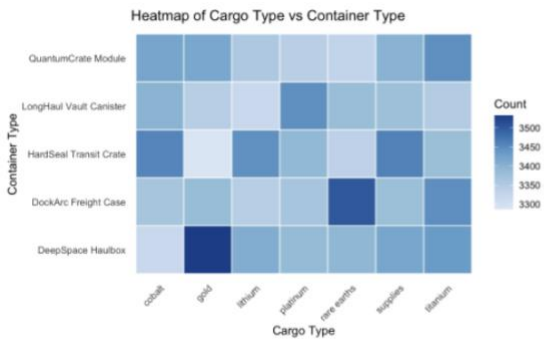


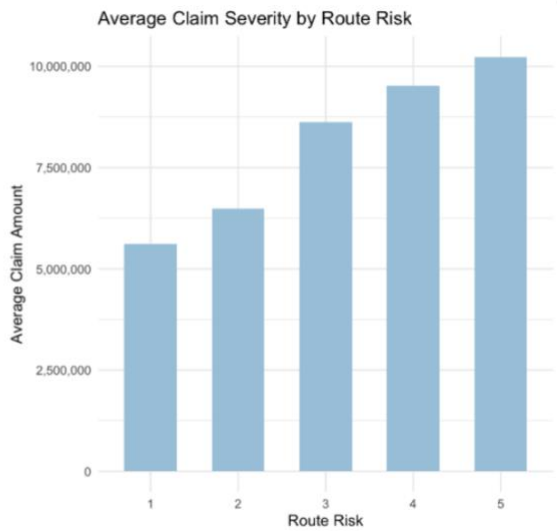
In the figure above, the average claim rate across all three solar systems, Epsilon, Helionis Cluster, and Zeta, demonstrates a clear and consistent negative relationship with exposure bin, declining steadily from approximately 0.75 at the lowest exposure bin [0.0, 0.1] to around 0.10 by the highest bin [0.9, 1]. This inverse trend suggests that entities with lower exposure levels exhibit higher claim rates, which may indicate that low-exposure periods correspond to concentrated or high-risk events rather than sustained, stable operations. A notable divergence is observed at the lowest exposure bin [0.0, 0.1], where Helionis Cluster records a considerably lower average claim rate of approximately 0.58 compared to Epsilon and Zeta which both sit closer to 0.75, suggesting that low-exposure entities within Helionis Cluster carry a relatively lower claim frequency than their counterparts in other systems. Beyond this initial gap, the three systems follow closely aligned trajectories across the remaining exposure bins, with no single system consistently dominating, and all three converging to similar claim rates of between 0.10 and 0.15 at the highest exposure levels. Overall, exposure bin emerges as a strong and consistent predictor of claim rate and should be considered an important variable in subsequent modelling.



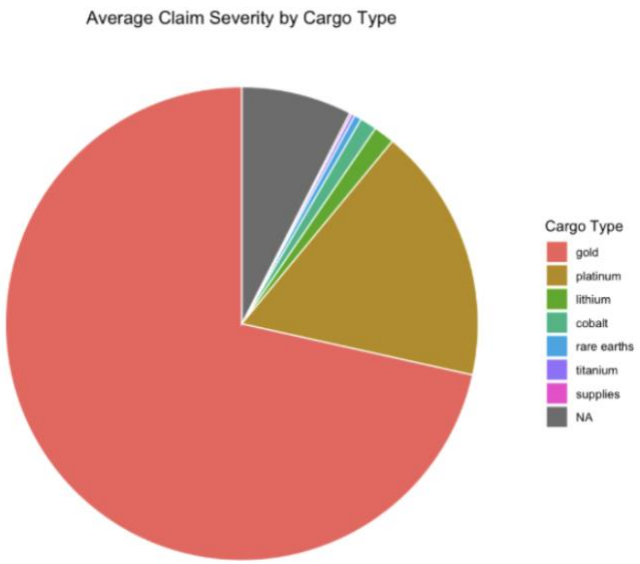
In the figure above, the heatmap of total claims by maintenance frequency reveals distinct differences in claim volumes across the three solar systems rather than across maintenance frequency levels. Epsilon consistently records the highest total claims across all maintenance frequencies, displayed through the darkest shading throughout its row, while Helionis Cluster falls in the mid-range and Zeta records the lowest total claims, remaining notably lighter across all frequencies. Importantly, maintenance frequency itself appears to have little discernible effect on total claims within any individual solar system, as the colour intensity remains largely uniform across all frequency levels from 0 to 6 within each row. This suggests that increasing maintenance frequency does not meaningfully reduce total claim volumes, which is a counterintuitive finding that may warrant further investigation in subsequent modelling. Overall, solar system emerges as a far stronger differentiator of total claims than maintenance frequency in this analysis.

Appendix E.2– Cargo Loss EDA

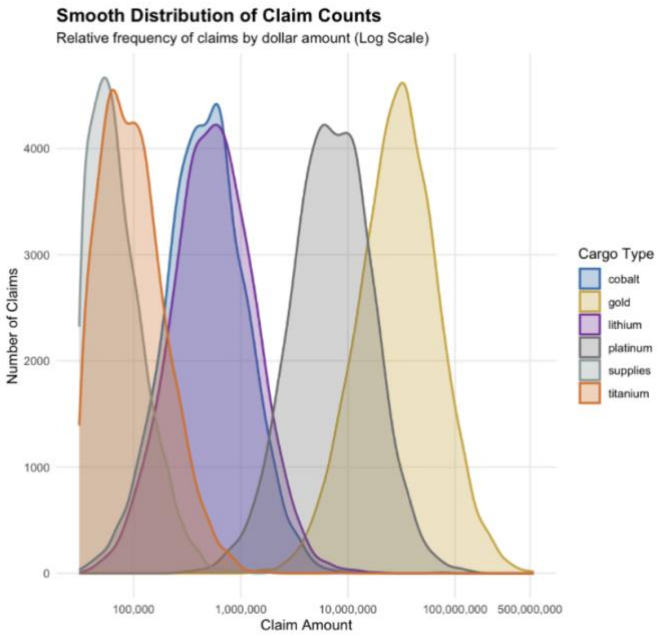




The data shows a direct correlation between route risk and claim outcomes. While lower-risk routes maintain a higher frequency of smaller claims, the average claim severity by route risk confirms that higher-risk corridors, likely those in the outer rim or zones with high interstellar interference, produce significantly larger individual losses. Furthermore, the heatmap of cargo vs. container type identifies specific logistical hotspots, suggesting that certain container configurations used for high-value minerals are more prone to recorded incidents.

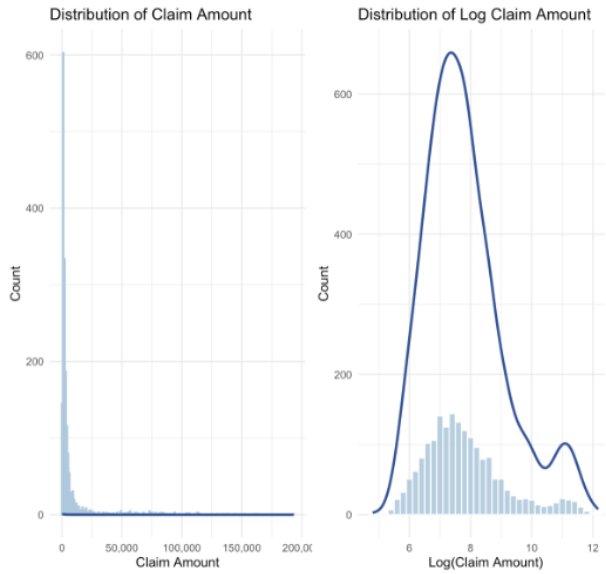


Gold and platinum act as extreme outliers in the portfolio, where gold accounts for an average severity of Đ42.19M per claim, with a maximum loss exceeding Đ542M. In contrast, Supplies and Titanium represent the high-frequency/low-severity bulk of the portfolio, with average claims under Đ135,000.

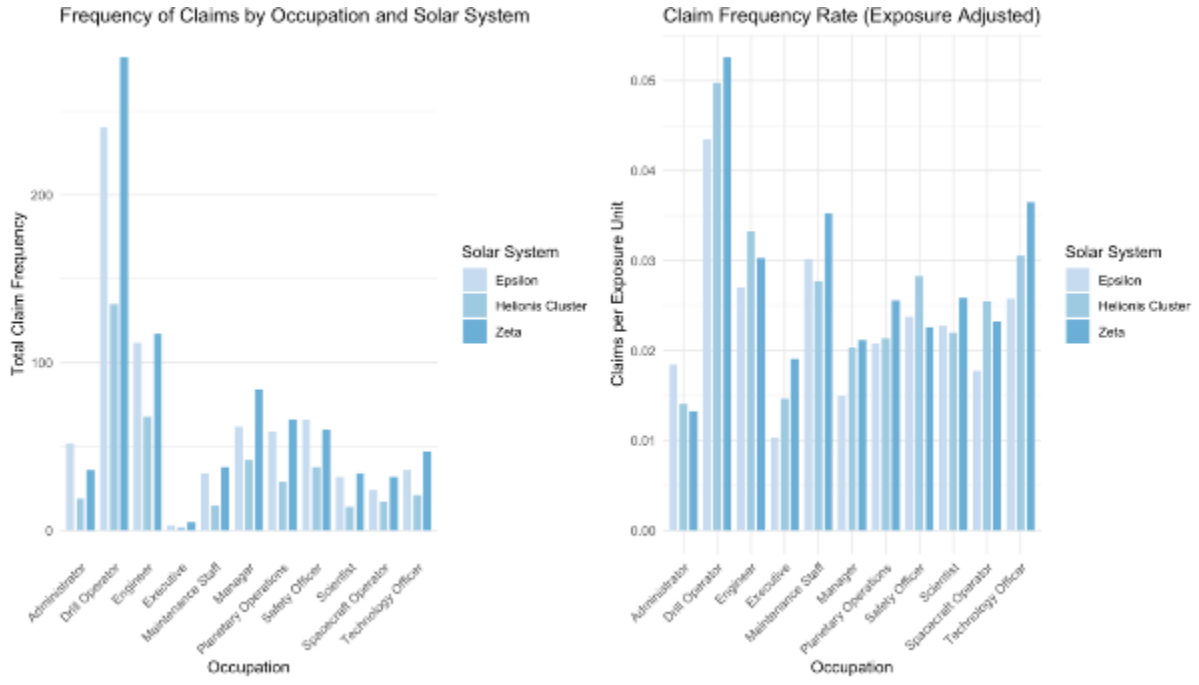


The Distribution of Claim Counts illustrates a wide, multi-modal spread in losses. While the density of claims for basic supplies is concentrated at the lower end of the financial spectrum, the distribution for precious metals like gold and platinum is shifted significantly to the right. This log-linear spread confirms that the portfolio is heavy-tailed, meaning a small percentage of high-value cargo claims drives the vast majority of the insurer's total capital exposure.

Appendix E.3– Workers' Compensation EDA

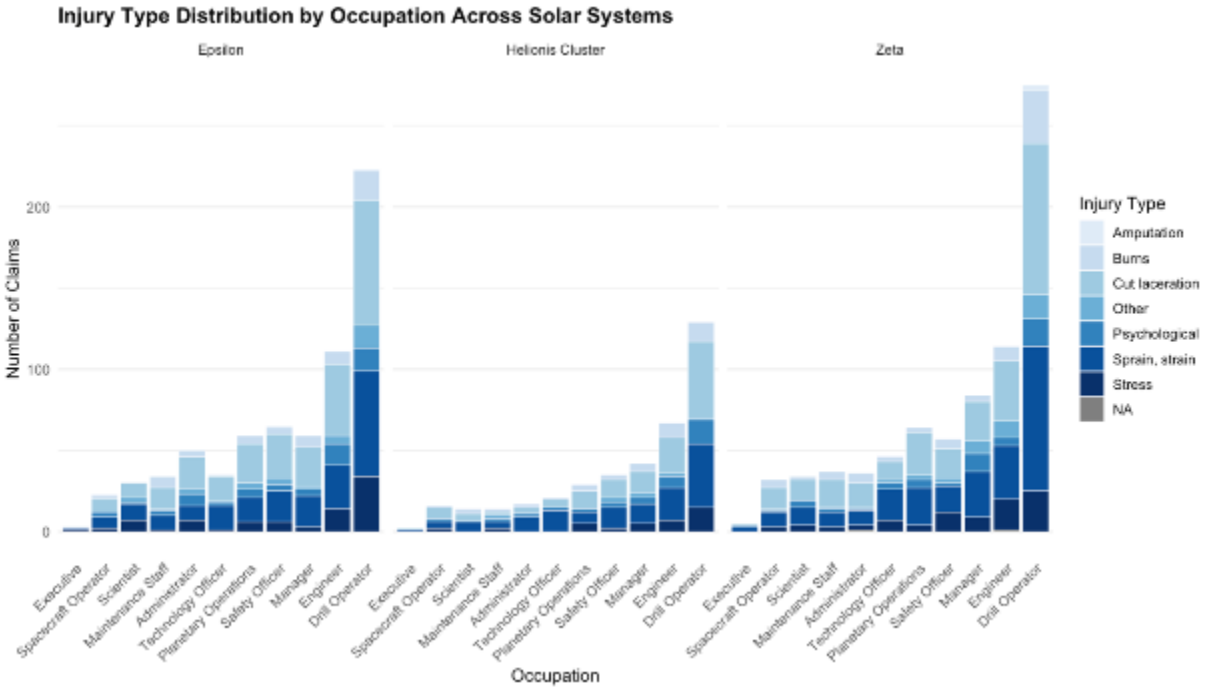


In the figure above, the distribution of claim amounts is substantially positively skewed, with most claims concentrated at lower values and a long right tail indicating the presence of rare but very large claims. The maximum claim amount recorded is Ð193,357.20, while the minimum is Ð120.76, highlighting a very wide range of claim severity. The presence of extreme outliers suggests that catastrophic claims occur and may significantly impact the overall cost of the portfolio. To better understand the distribution, a log transformation was applied to the claim amounts, which produces a more symmetric distribution. However, even after transformation, the data still exhibits inherent characteristics of positive skewness, confirming the heavy-tailed nature of claim severity.



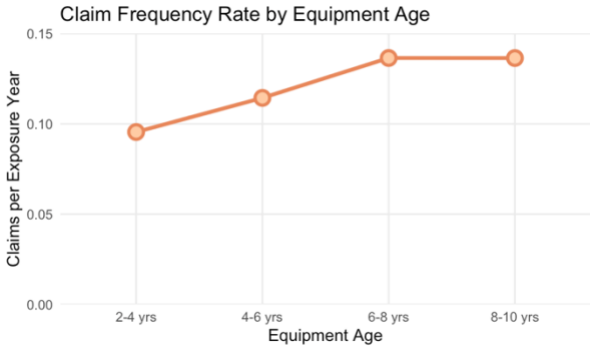
From the figure on the left-hand side, it is evident that the number of claims across occupations largely reflects the workforce distribution. Drill operators experience the highest number of claims in all three solar systems, while executive roles report the fewest claims. This pattern is expected given the physically demanding nature of drilling work and the larger workforce employed in this occupation. In contrast, executives typically perform less physically intensive tasks and represent a smaller portion of the workforce, which contributes to their lower claim counts.

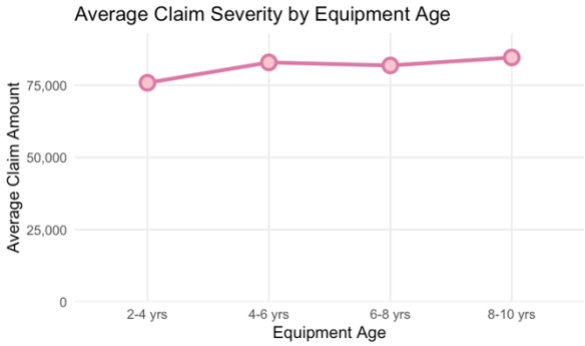
Adjusting claim counts for exposure (shown in the right-hand figure) accounts for time or level of risk each worker is exposed to potential injury. This provides a more accurate measure of occupational risk, as it reflects not just the total number of claims, but the frequency of claims relative to the opportunities for them to occur. Although drill operators still exhibit the highest claim frequency, the exposure-adjusted results reveal additional insights. While the Helionis Cluster previously appeared to have fewer claims due to its smaller workforce, exposure adjustment shows that it experiences relatively higher claim rates. This suggests that working conditions in the Helionis Cluster may be more hazardous, resulting in a higher risk of injury per worker. Some occupations also show higher claim frequency per unit of exposure, indicating riskier work environments.



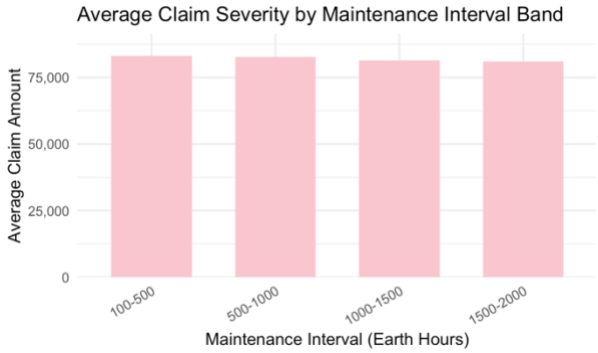
Injury patterns also differ across solar systems. Looking at the comparative plots above, for drill operators, cut or laceration injuries and sprains or strains remain the most common injury types across all locations. However, the number of claims associated with these injuries is significantly higher in the Zeta solar system compared to the others, while the Helionis Cluster records the lowest counts. Stress-related injuries are more prominent in the Epsilon solar system, which may suggest higher workplace pressure or insufficient worker support. These differences indicate that environmental or operational factors within each solar system may influence injury risk.

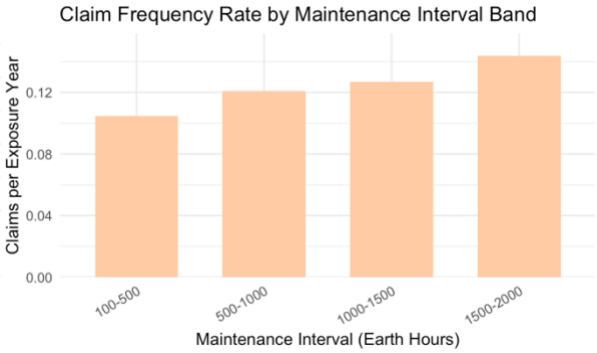
Appendix E.4– Equipment Failure EDA



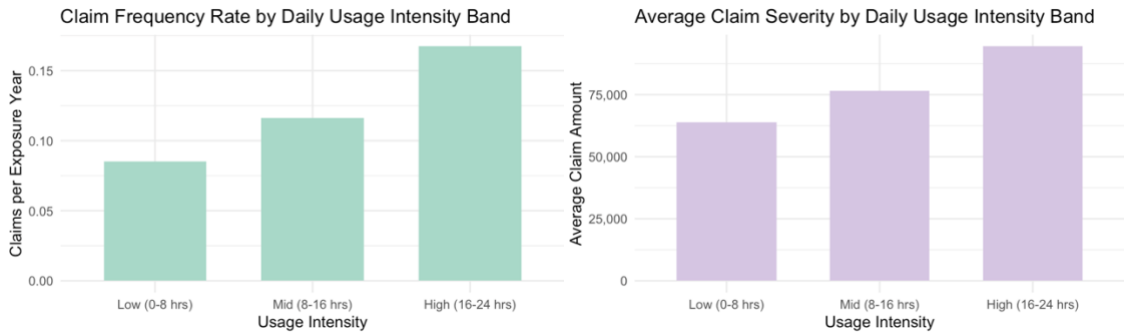


The "Claim Frequency Rate by Equipment Age" plot depicts how claim frequency rises steadily through mid-life before plateauing in the 6–10 year range, exemplifying the accumulation of mechanical degradation over time. However, severity does not mirror this pattern as it remains relatively stable across all age bands with only a modest upward drift in the oldest equipment, suggesting that when failures occur they tend to be similar in nature regardless of age. The real risk amplification with ageing equipment therefore comes from claims happening more often rather than becoming more expensive when they do.

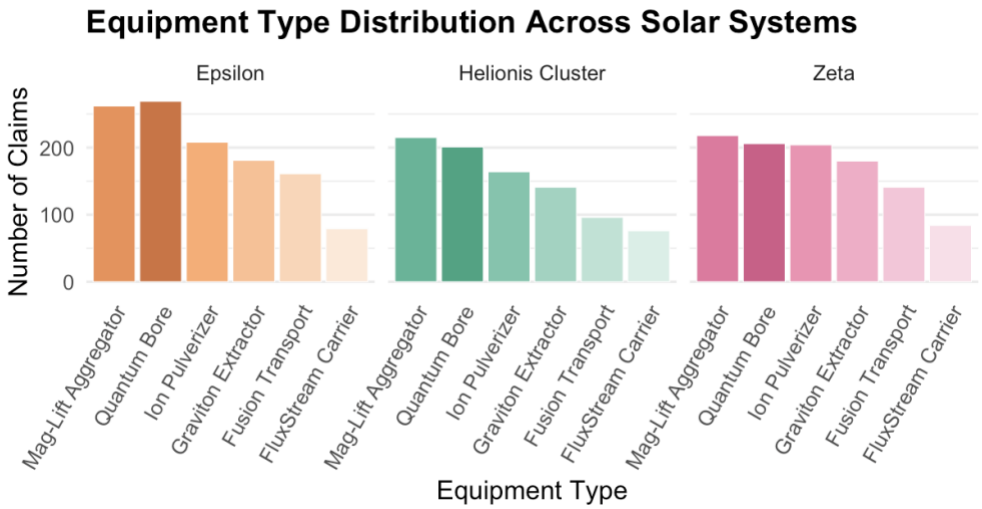




The figures above showcase how equipment on longer service cycles claims noticeably more frequently, with a consistent step-up across each maintenance band from the tightest to the loosest intervals. However, severity tells a completely different story, remaining almost entirely flat across all maintenance bands. This divergence reveals that stretching service intervals increases how often equipment breaks down but has no meaningful effect on the cost or severity of damage when it does, pointing to maintenance as a pure frequency driver rather than a severity one.



Both frequency and severity escalate sharply as daily operating hours increase, but the pattern is more pronounced in severity than in any other variable examined. Single-shift equipment claims relatively infrequently and at the lowest average costs, while double-shift operation pushes both metrics meaningfully higher. Near-continuous operation produces by far the steepest jump as claim rates more than double compared to low-intensity use, and average claim costs climb substantially above the mid-intensity band. This suggests that under continuous stress, components fail more catastrophically rather than wearing gradually, making high-utilisation equipment distinctly more damaging across both dimensions simultaneously.



The claim distribution across equipment types is strikingly consistent across all three solar systems with Epsilon, Helionis Cluster, and Zeta all show the same descending hierarchy. Mag-Lift Aggregators and Quantum Bores generate the highest claim volumes, followed by Ion Pulverizers and Graviton Extractors in the middle tier, while Fusion Transports and FluxStream Carriers sit notably lower. This uniformity across geographically distinct systems suggests the claim pattern is driven by equipment-intrinsic characteristics rather than any system-specific conditions. Epsilon appears to carry the highest absolute claim counts overall, particularly for its top two equipment types, while Helionis Cluster consistently runs the lowest overall, hinting at either better operating conditions, lower fleet exposure, or more disciplined maintenance culture within that system.

Appendix F – Correlated Risk Scenario

The following table highlights the full results for the mean annual loss, Value-at-Risk, and tail Value-at-Risk for 1-in-40, and 1-in-200 year events for each hazard.

Correlated Risk Scenarios Key Results Summary	
Metric	**Value (Đ)**
INDIVIDUAL LINE METRICS	
Cargo Loss (ex. Gold + Platinum)	
Mean Annual Loss	**15,382,309,139**
VaR 1-in-40 (97.5th pct)	**15,770,004,862**
TVaR 1-in-40 (97.5th pct)	**15,849,122,190**
VaR 1-in-200 (99.5th pct)	**15,902,873,391**
TVaR 1-in-200 (99.5th pct)	**15,977,520,615**
Business Interruption	
Mean Annual Loss	**2,652,889,925**
VaR 1-in-40 (97.5th pct)	**6,848,717,261**
TVaR 1-in-40 (97.5th pct)	**8,899,808,285**
VaR 1-in-200 (99.5th pct)	**10,161,960,434**
TVaR 1-in-200 (99.5th pct)	**12,068,672,744**
Workers Comp	
Mean Annual Loss	**11,393,240**
VaR 1-in-40 (97.5th pct)	**42,797,822**
TVaR 1-in-40 (97.5th pct)	**53,219,945**
VaR 1-in-200 (99.5th pct)	**59,215,433**
TVaR 1-in-200 (99.5th pct)	**69,359,368**
Equipment Failure	
Mean Annual Loss	**253,976,329**
VaR 1-in-40 (97.5th pct)	**368,996,833**
TVaR 1-in-40 (97.5th pct)	**393,736,588**
VaR 1-in-200 (99.5th pct)	**407,917,532**
TVaR 1-in-200 (99.5th pct)	**427,687,597**
PORTFOLIO TOTALS — KEY CORRELATION SCENARIOS	
rho = -0.5 (Diversifying (negative corr.))	
Portfolio Mean	**17,623,640,029**

VaR 1-in-40 (97.5th)	**23,277,390,522**
TVaR 1-in-40 (97.5th)	**25,390,397,745**
VaR 1-in-200 (99.5th)	**26,729,480,446**
TVaR 1-in-200 (99.5th)	**28,814,685,201**
rho = -0.2 (Mildly diversifying)	
Portfolio Mean	**17,546,156,801**
VaR 1-in-40 (97.5th)	**22,763,244,790**
TVaR 1-in-40 (97.5th)	**24,724,899,367**
VaR 1-in-200 (99.5th)	**26,087,201,282**
TVaR 1-in-200 (99.5th)	**27,663,177,165**
rho = -0.1 (Near independent (slight −))	
Portfolio Mean	**17,528,434,956**
VaR 1-in-40 (97.5th)	**22,614,143,604**
TVaR 1-in-40 (97.5th)	**24,593,464,765**
VaR 1-in-200 (99.5th)	**25,858,604,308**
TVaR 1-in-200 (99.5th)	**27,443,643,171**
rho = +0.0 (Independence (baseline))	
Portfolio Mean	**17,526,860,070**
VaR 1-in-40 (97.5th)	**22,499,979,429**
TVaR 1-in-40 (97.5th)	**24,556,556,764**
VaR 1-in-200 (99.5th)	**25,898,825,106**
TVaR 1-in-200 (99.5th)	**27,776,389,322**
rho = +0.1 (Mild positive correlation)	
Portfolio Mean	**17,525,693,769**
VaR 1-in-40 (97.5th)	**22,726,824,099**
TVaR 1-in-40 (97.5th)	**24,937,724,833**
VaR 1-in-200 (99.5th)	**26,302,585,576**
TVaR 1-in-200 (99.5th)	**28,281,929,275**
rho = +0.2 (Moderate positive correlation)	
Portfolio Mean	**17,498,442,965**
VaR 1-in-40 (97.5th)	**22,518,047,720**
TVaR 1-in-40 (97.5th)	**24,527,712,553**
VaR 1-in-200 (99.5th)	**25,781,080,679**
TVaR 1-in-200 (99.5th)	**27,912,337,063**

rho = +0.5 (High correlation — Great Flare scenario)	
Portfolio Mean	**17,532,817,477**
VaR 1-in-40 (97.5th)	**22,857,577,806**
TVaR 1-in-40 (97.5th)	**24,905,507,403**
VaR 1-in-200 (99.5th)	**26,428,172,551**
TVaR 1-in-200 (99.5th)	**28,029,881,034**

Appendix G – Threat Table

Rank	Threat Name	Primary System(s)	Impact Level	Likelihood	Rationale
1	Solar Storm / Correlated Flare	All Systems	Extreme	Low	Drives the Ð 42B+ loss in catastrophic modeling; affects all habitats simultaneously.
2	Orbital Shear / Gravity Gradient	Oryn Delta	High	Moderate	Asymmetric asteroid ring causes localized "crushing" events and high-severity equipment loss.
3	Radiation Spike / Binary Flare	Bayesia	Medium	High	Frequent K-type binary alignments cause "attritional" claims (high frequency, lower severity per claim).
4	Mechanical/Atmospheric Erosion	Epsilon / Helionis	Medium	Moderate	Corrosive atmospheres (Epsilon) and dust (Helionis) lead to steady frequency of maintenance claims.
5	Orbital Saturation (Debris)	All Systems	Low	Increasing	Reflects the 2174 legal settlement (Ð 40M); manageable but poses a long-term liability risk.
6	Fatigue / Extended Hours Worked	All Systems	Medium	Moderate – High	Increases accident frequency, reflected in claim lengths
7	Climate Risk	All Systems	Extreme	Low – Moderate	Long-term environmental instability (solar cycles, asteroid drift, thermal fluctuations)

					disrupts infrastructure and increase systemic loss potential across multiple hazards
8	Inflation Risk	All systems	Moderate – High	Moderate	Rising repair, replacement, and wage costs increase claim severity, particularly for equipment replacement and workers' compensation liabilities

Appendix H – Secondary Assumptions

Several supplementary assumptions simplify modelling and computational efficiency but have a lower impact on pricing or capital outcomes. Parameter uncertainty is not explicitly modelled, and claims are assumed to settle within the same financial period, avoiding long-tail reserving dynamics. Macroeconomic variables are projected using ARIMA models based on historical trends, and coverage rules, deductible tiers, and fleet discounts are applied consistently across the portfolio. Compound aggregate losses are approximated using a Gamma convolution, with a GPD tail fit applied above the 95th percentile for extreme severity assessment. Frequency–severity dependence within cargo is modelled via vine copulas, with Gaussian fallbacks if convergence fails. Scenario horizons are fixed at 3 and 10 years, and investment returns on reserves are not modelled. These assumptions are documented for completeness and transparency but are expected to have minimal impact on overall premium levels, reserve adequacy, or tail risk estimates.

Appendix I – Data and Data Limitations

Data Sources

Provided and used data sources have been referenced below

Source	Description	Used For
srcsc-2026-claims-cargo.xlsx	Provided project data: CL claims frequency and severity	Cargo GLM fitting (NB frequency, Gamma severity), EDA
srcsc-2026-claims-business-interruption.xlsx	Provided project data: BI claims frequency and severity	BI GLM fitting, mu/theta/shape/scale parameter extraction
srcsc-2026-claims-workers-comp.xlsx	Provided project data: WC claims frequency and severity	WC NB frequency, Gamma severity, anchor calibration
srcsc-2026-claims-equipment-failure.xlsx	Provided project data: EF claims frequency and severity	EF Poisson frequency, Gamma severity, anchor calibration

srcsc-2026-cosmic-quarry-inventory.xlsx	Provided project data: equipment inventory by system (~505 units)	EF exposure base (unit count for lambda derivation)
srcsc-2026-cosmic-quarry-personnel.xlsx	Provided project data: personnel headcount by system (~86,765 workers)	WC exposure base (worker count for frequency per exposure)

Data Limitations

Some further Data Limitations regarding hazard coverage datasets have been summarised below.

Limitation	Description	Impact
No gold/platinum severity in modelling dataset	Once excluded, only 18,673 severity records remain against 85,332 frequency records, which is a 4.6:1 ratio, meaning the majority of frequency observations have no matched severity record	Severity model trained on a non-random subsample of claims; the Gamma GLM may underweight low-value claims if small claims were less likely to have full severity data recorded. Mitigated by anchoring sev_scale to mean(claim_amount)/sev_shape.
No multi-system claimtagging	Claims data does not flag which solar system each claim originated from	System-specific pricing relies on route_risk, debris_density, and solar_radiation GLM coefficients as proxies for system membership rather than direct system identifiers. Oryn Delta's tail risk may be underestimated if its claims are diluted in a pooled dataset.
Historical data spans Epsilon & Zeta systems	Calibration dataset includes two systems not in scope for the RFP	Parameters (especially frequency intercepts) may be diluted by operating conditions in Epsilon and Zeta that differ materially from Helionis/Bayesia/Oryn Delta. We accept this as a conservative bias, as if Epsilon/Zeta have lower risk profiles, our estimates are slightly conservative.
No time-series dimension in claims data	Data is cross-sectional; no year-of-occurrence field is visible	Cannot model claims development, trend, or latent demand surge. Inflation loading of up to 40% stress-tested separately (Section 3b) to compensate.

Appendix J – References

Allianz. "Understanding and Mitigating the Impacts of Space-Related Risks." *Understanding and Mitigating the Impacts of Space-Related Risks*, 2025, commercial.allianz.com/content/dam/onemarketing/commercial/commercial/reports/commercial-cro-eri-space-risk.pdf.

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